CSA15_CO2...lem Solving

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I _K_Nandha kumar k (1921260223_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __K Nandha kumar__

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded

:

CO2 - ASSESSMENT

TOOL - 1

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course code : CSA1519

course name : cloud computing

1. Compare Type - 1 and Type - 2 Hypervisors :

	Type - 1 Hypervison	Type - 2 Hypervizon
*	Runs directly on physical Hardcoare.	Run on top of a host operating system.
*.	Provides High Penformance with low overhead.	Has more overhead and the comparatively lown perfomance
*.	Offers strong Isolation and secwicty	Easy to Install and manage.
*	Suitable for mission - critical cloud applications.	Suitable for tatting and Development.
*.	Eg: VMare ESXi, xen	Eg: Virtual Box, VMwore workstatios

2. Virtualization Isolation and Attack surfaces :

*. Runs financo, healthare and studard portals in seperate VMs.

*. Allocate separate CPU, memory, storage and network resources.

*. Use virtual network and access controls.

*. Applies encryption to peract sensitive information.

a. Two major attack surfaces:

* Hypervisor Vulnerabilities.
* Virtual network attacks.

3. VM Host Utilization Analysis:

* CPU 92 %.
  Memory 88 %.
  Storage I/O 35 ms
  VM Boot time 170 seconds

* CPU 92 % → High utilization causes processing delays
* Memory 88 % → memory pressure may cause swapping and slow performance.
* Storage latency 35 ms → slow disk operations delay VM activities
* Boot time 170s → indicates possible resource contention or slow storage
* Corrective actions → Add CPU and RAM, use faster SSD storage, balance workloads, and optimize VM allocation.

4. Software Defined Datacenter (SDDC):

* SDDC manages compute, storage, and networking through software.
* compute virtualization → creates and manages VMs dynamically
* Storage virtualization → Combine storage resources into a flexible storage pool.
* Agility → Enable farm providing, automation and resource scaling.

∴ SDDC is more flexible, scalable and cost-efficient than a traditional datacenter.

5. Multi-cloud Federation and Identity Mismatch:

* Different cloud providers use different identity formats and authentication methods.

* Establishes trusted authentication blw cloud providers.

* single sign-on allows users to access multiple cloud services with one login.

* Use SAML or openID connect (OIDC) for federated authentication.

* Apply encryption, least privilage access and data minization.